6th Theory of Advanced Machine Learning

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Course Guide

- Students may take either Deep Reinforcement Learning (Part 1) or Generative Adversarial Networks (Part 2).
- **Deep Reinforcement Learning:**
  - It combines deep learning and reinforcement learning and is used to operate robots and games.
- **Generative Adversarial Networks (GANs):**
  - GANs are used for tasks such as data augmentation and generating high-quality images. They consist of two networks: a generator and a discriminator. For example, when generating images, the generator outputs an image while the discriminator determines its authenticity. The generator learns to deceive the discriminator, while the discriminator learns to identify images more accurately.

https://ja.wikipedia.org/wiki/%E6%95%B5%E5%AF%BE%E7%9A%84%E7%94%9F%E6%88%90%E3%83%8D%E3%83%83%E3%83%88%E3%83%AF%E3%83%BC%E3%82%AF

(Part1) Deep Reinforcement Learning

You should watch all youtube videos about **deep reinforcement learning** below.

- MIT 6.S091：Deep RL
  https://www.youtube.com/watch?v=zR11FLZ-O9M　1 hour
- MIT 6.S191（2019）： deep reinforcement learning
  https://www.youtube.com/watch?v=i6Mi2_QM3rA　45 minutes
- CS 285: Lecture 1, Part 1　deep reinforcement learning
  https://www.youtube.com/watch?v=JHrlF10v2Og&list=PL_iWQOsE6TfURIIhCrlt-wj9ByIVpbfGc　10 minutes
- CS 285: Lecture 1, Part 2　deep reinforcement learning
  https://www.youtube.com/watch?v=IoF7D0qec0I&list=PL_iWQOsE6TfURIIhCrlt-wj9ByIVpbfGc&index=2　12 minutes

(Part2) Generative Adversarial Netrowks

You should watch all youtube videos about **GANs** below.

- **What are GANs (Generative Adversarial Networks)?**

https://www.youtube.com/watch?v=TpMIssRdhco 8 minutes

- **Generative Adversarial Networks (GANs) – Computerphile**

https://www.youtube.com/watch?v=Sw9r8CL98N0 21 minutes

- **Very good Introduction to GANs**

https://www.youtube.com/watch?v=9JpdAg6uMXs 31 minutes

- **NIPS 2016 - Generative Adversarial Networks - Ian Goodfellow**

For students to understand in depth

https://www.youtube.com/watch?v=AJVyzd0rqdc 2 hours

・**Mathematical foundation of GANs**

https://www.youtube.com/watch?v=Gib_kiXgnvA 17 minutes